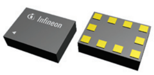
It provides ultra-low latency and low-power bandwidth scalability to CPUs and specialised accelerator devices. Now high-speed, low-latency connectivity can be easily achieved for distributed, heterogeneous compute systems

*Microchip Technology*
*www.microchip.com*

## Antenna tuners

The new range of innovative antenna tuners, including BGSA144ML10, BGSA400ML10, BGSA403ML10, and




BGSA14M2N10, are suitable for sub-7.2GHz New Radio applications, supporting both 4G and 5G in smartphones, notebooks, wearables, VR headsets, smart homes, and other cellular applications. RF engineers can now face the challenge of complex antenna design brought by increasing connected devices and introduction of 5G.

*Infineon Technologies*
*www.infineon.com*

## General-purpose microprocessor

The RZ/Five general-purpose microprocessor unit (MPU) built around a 64-bit RISC-V CPU core employs the


Andes AX45MP, based on the RISC-V CPU instruction set architecture (ISA). It augments the company's previously available Arm CPU core based MPUs, expanding customer options and providing more flexibility in the product development process. RZ/Five is optimised to provide the performance and peripheral functions required of IoT endpoint devices. It comes in a small, compact package to address less complex designs more efficiently.

*Renesas Electronics*
*www.renesas.com*

## Depth image sensor

VD55H1 is the first of a series of high-resolution time-of-flight sensors that bring advanced 3D depth imaging to smartphones and other devices. The sensor maps three-dimensional

surfaces by measuring the distance to over half a million points. Objects can be detected up to five metres from the sensor, and even further with patterned illumination. It addresses emerging AR/VR market use cases, including room mapping, gaming, and 3D avatars. In smartphones, the sensor enhances the performance of camera-system features, including bokeh effect, multi-camera selection, and video segmentation. Face-authentication security is also improved with higher resolution and more accurate 3D images. In robotics, the VD55H1 provides high-fidelity 3D scene mapping for all target distances.

*STMicroelectronics*
*www.st.com*

## Diode for system protection

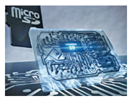
The DZDH0401DW is a diode controller for system protection against reverse discharge. Emulating an


ideal diode by driving a P-channel MOSFET, the controller provides the high-side rail isolation necessary for reverse current blocking in systems up to 40V. The compact size and high performance capabilities of this arrangement mean that it is optimised for use in cordless power tools and autonomous household/garden appliances, such as smart vacuum cleaners and smart lawnmowers. It is also well-suited for other applications, including OR-ing rectifiers used in the hot-swappable and N + 1 redundancy power supplies used in data centre servers.

*DIODES Incorporated*
*www.diodes.com*

## NAND flash controllers

The new S9 family of secure digital (SD) and microSD memory card controllers provide reliable NAND flash controllers for meeting the

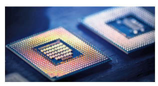


requirements of demanding applications with turnkey firmware. The controller's FlashXE ECC and hyReliability features guarantee extended endurance, data integrity, and power-fail safety coping with demanding applications in markets such as industrial automation, telecom, networking, and medical equipment. With the hyMap flash translation layer, the S9 achieves minimal write amplification and maximum endurance, which enables an efficient use of 3D NAND flash, resulting in lowest field failures over time.

*Hyperstone*
*www.hyperstone.com*

## Gluon coprocessor for edge

The Gluon coprocessor for edge AI inference claims to deliver competitive



performance compared to Nvidia in vision workloads such as object detection. Gluon is based on an instruction set architecture (ISA) optimised for AI. The ISA refers to the instructions that a chip can process.